

CART470 – Distributed Listening

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Project Focus

In collaboration with Professor Gabriel Vigliensoni, we are building an open-source distributed listening system for his CART 346 Digital Sound class that uses mobile phones as speaker nodes for spatialized sound installations. Upon entering a room, participants scan a QR code to load a web app in their phone's browser, then place their phone on the ground or a table. Together, the phones act as a spatialized speaker system that can be set up in any room or layout without dedicated audio hardware.

The project is meant to be a skeleton, or base that others can build on. Students in Gabriel's CART 346 Digital Sound class will be able to fork our GitHub repository, add their own audio files, adjust the UI, and write a JSON score that dictates which sounds play on which phones and when.

Thesis/Project Outcome

Our goal/thesis is to build a real-time, web-based system that distributes audio across a group of mobile phones in near synchronicity. As participants join by scanning a QR code, the system keeps track of each phone and assigns it to a speaker group based on the piece's JSON score. During playback, sound moves between these groups according to how the sound artist has programmed it, turning the room into a spatialized sound installation.

The intended outcome of the project is to create an adaptable, open-source tool for artists. By forking our repository and writing their own score, students/artists will be able to present distributed, spatial pieces. The final outcome will be a working demo hosted on Compute Canada, a public GitHub repository, a template score and documentation that make the system easy for others to work with and build on.

Knowledge Base/Fields/Domains/Keywords

Audiovisual installation, Open Source, WebSockets, Web hosting, JavaScript, Reactive UI/UX, Distributed Audio

Link to team Kanban

<https://app.fizzy.do/6280767/boards/03gvol14dsy6pof4ziyx2igqt>

Learning Objectives

1. Learn how to use WebSockets
2. Learn how to use the Web Audio API to load and play audio on phones
3. Learn how to keep audio in sync across many devices
4. Design a JSON score template that is simple enough for students to use
5. Host the server through Compute Canada
6. Write a clear, concise codebase with documentation so others can fork and use the project
7. Learn to work in a group and with client expectations

Learning Activities

- Research on using the Web Audio API
- Understand how to host the server using Compute Canada for the project
- Create an example JSON template needed for the project
- Learn to make a reactive UI using JavaScript and JSON

MileStones

1. Proof of concept: Have it work locally with a loaded music file, and have the server. When users connect to the page, all connected phones will play a default test sound.
2. Have the project be able to play audio on different groups of phones according to JSON file instructions. Apply feedback from Gabriel.
3. Be able to host the server using Compute Canada. UI dynamically reacts according to audio.

4. Have a server on Compute Canada that can split audio files to multiple phones. The controller will create the number of grouped phones and play the audio

Weekly Plan

Week#	Topic	Detail	Work Due
Sep 23	Define project	Define scope, way of getting audio, define main function	Sep 30
Sep 30	Set up server and questions for Gabriel + Create rough UI wireframe	Write more questions to ask Gabriel in the Discord for the Friday meeting	Oct 7
Oct 7	Iteration #1	Have it work locally with a music file, and test. When users connect to the page, all connected phones will play a default test sound.	Oct 14
Oct 14	Gather feedback	Test out and gather feedback from Gabriel	Oct 21
Oct 21	Iteration #2 Prototype and presentation due (Midterm due next week)	Have the project be able to play different files of audio on different groups of phones according to JSON	Oct 28
Oct 28	Finalize dynamic UI and implement front-end	Design and implement the front end	Nov 4
Nov 4	User testing and feedback	Improve UI and gather feedback from stakeholders	Nov 11
Nov 11	Iteration #3	Have the project hosted on the Compute Canada server+UI	Nov 18

		dynamically reacts according to audio.	
Nov 18	QA testing	Test out project in different scenarios to find bugs and edge cases	Nov 25
Nov 25	Gather feedback	Get feedback from Gabriel	Dec 1
Dec 1	Iteration #4	Finalize bugs and implement final feedback	Dec 9
Dec 9	Finalization	Finalize last-minute bugs and implement details	Dec ?